

# FINAL EXPLANATION: BIOLOGY GRADE 11

*Teacher Key: Exemplar Answers and Marking Guide*

## FINAL EXPLANATION: BIOLOGY GRADE 11: TAXONOMY I (SUB-STRAND 1.1)

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation. It answers the Driving Question: How can we organise the many different living organisms around us so that everyone can identify and study them in the same way? Think back to Lesson 1, when your group sorted the goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito, and found that other groups sorted them differently and used different Kikuyu, Luo, Kiswahili and English names. Answer each section in your own words in the space under the prompt. Use evidence from the activities you did, including the cards, videos, charts and the Final Classification Challenge. Where a prompt asks you to, give a Claim, Evidence and Reasoning. Write scientific names correctly: genus with a capital letter, species with small letters, and italicised or underlined.

### Part 1: Why did our groupings differ, and why is that a problem?

**In Lesson 1 you sorted eight familiar organisms by your own rules, and other groups sorted them differently. (a) Describe two different ways groups sorted the organisms. (b) Why could every group say it was correct? (c) Why did this still cause a problem when people tried to identify, study and talk about organisms?**

(a) One group sorted by where the organisms live (for example, land versus water). Another sorted by usefulness (food, pests, disease), and another by whether the organism is green or can move. (b) Each group was correct by its own rule, because there is no single right criterion for sorting when everyone picks a different one. (c) Because the rules differed, the same organism ended up in different groups. The same organism also has different local names in Kikuyu, Luo, Kiswahili and English, so two people may not be sure they are talking about the same organism. Local names and personal rules are useful at home but cannot be used by everyone in the same way. We need a universal, agreed system of classification. This is the science of taxonomy, which means naming, describing and classifying organisms.

### Part 2: Why do scientists classify organisms?

**Think about the case studies you investigated in Lesson 2. Write a Claim, Evidence and Reasoning explaining why classifying organisms and identifying them correctly matters. Include one reason that links to conserving living things.**

Claim: Classification is needed because there are so many different organisms that we cannot study or identify them one by one without an organised system, and wrong identification can cause harm. Evidence: In the cases we studied, an edible mushroom can be confused with a poisonous one, and wrongly identifying it can be dangerous. Among insects, some are useful while others are harmful, so we must be able to tell them apart by their features. Among plants, crops such as maize and beans must be told apart from weeds such as couch grass and black jack. Reasoning: Correct identification depends on checking shared features, not on local names alone, which differ between communities. A shared system lets people identify, study and communicate about organisms accurately, for example to recognise a mosquito that spreads disease or a crop pest. It also helps conservation, because we must know which organisms exist and how they are related in order to protect biodiversity.

### Part 3: Taxonomic units and scientific names

**(a) List the seven taxonomic ranks in order from the broadest to the most specific.**  
**(b) Explain what happens to the number of organisms in a group and how similar they are as you move down the ranks.**  
**(c) Explain how a scientific name is written and why a two-part scientific name solves the problem of different local names. Use maize (*Zea mays*) as an example.**

(a) Kingdom, Phylum (Division in plants), Class, Order, Family, Genus, Species. (b) The ranks are nested, each one inside the one above it. As you move down, there are fewer organisms in each group and they share more features, so they are more similar to each other. Species is the most specific rank. (c) Binomial nomenclature gives every organism a two-part Latin name: the genus first with a capital letter, then the species with small letters, with each part in italics or underlined separately, for example *\*Zea mays\** (or Zea mays when handwritten). Maize is called 'mahindi' in Kiswahili and has other names in other languages, but it has one agreed scientific name that is the same worldwide. A Luo farmer in Kisumu and a Kikuyu trader in Nairobi can therefore be sure they mean the same organism.

### Part 4: Classifying into the five kingdoms (Final Classification Challenge)

**Choose any four of these organisms: *Spirogyra*, *Escherichia coli*, *Rhizopus* on stale bread, *Agaricus* (mushroom), *Zea mays*, *Tilapia* (*Oreochromis niloticus*), *Anopheles* mosquito, *Amoeba*, *Acacia*. For each one, name its kingdom (choose from Monera, Protocista, Fungi, Plantae, Animalia) and give the observable evidence you used. Then write one sentence explaining why these characteristics are better for grouping than colour, habitat or usefulness.**

Any four of these are acceptable. *Escherichia coli*: Monera, because it is a bacterium with a simple cell without a true nucleus and it is single-celled. *Amoeba*: Protocista, because it is a single-celled organism with a nucleus that feeds by taking in food. *Spirogyra*: Protocista, because it is an alga that has chlorophyll and makes its own food and is found in water such as the shores of Lake Victoria. *Rhizopus*: Fungi, because it has cell walls and feeds by absorbing food from dead matter such as stale mandazi or bread, and has no chlorophyll. *Agaricus*: Fungi, for the same reasons. *Zea mays* and *Acacia*: Plantae, because they are many-celled, have cell walls and chlorophyll, and make their own food. *Tilapia* and *Anopheles* mosquito: Animalia, because they are many-celled, have no cell walls or chlorophyll, and feed on other organisms. Characteristics such as cell type, number of cells, cell wall, chlorophyll and mode of nutrition are shared features that anyone can check in the same way, while colour, habitat or usefulness change with the person and give different groups.

### Part 5: Putting it all together

**Compare how your group sorted the eight organisms in Lesson 1 with how you would now classify and name them. Write a final explanation that**

In Lesson 1 we sorted the goat, chicken, maize plant, mushroom, algae, bacteria, bean plant and mosquito by habitat, colour, size or usefulness, and each group got different results and used different local names. Now we know that living organisms are classified because there is a huge diversity of them, and taxonomy gives a scientific way of identifying, naming and grouping them by their similarities. Organisms are arranged in a hierarchy from Kingdom to Species, with each level more specific. Using cell structure, organisation and mode of nutrition, we place them in the five kingdoms: Monera (bacteria), Protocista (algae), Fungi

|                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>answers the Driving Question: How can living organisms be systematically organised, named and identified? Your explanation should connect classification, the hierarchy of ranks, scientific names and the five kingdoms, and say why taxonomy is important to people in Kenya.</b> | (mushroom), Plantae (maize, bean plant) and Animalia (goat, chicken, mosquito). Each species also gets a universal two-part scientific name, for example <i>*Zea mays*</i> or <i>*Capra hircus*</i> , written with a capital genus and a lower-case species, in italics or underlined. Because everyone uses the same evidence and names, people in Nairobi, Kisumu and Mombasa and across the world can identify, study and communicate about the same organism in the same way. This is important for recognising disease vectors such as the Anopheles mosquito, crop pests and useful organisms, and for conserving biodiversity. |
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| RUBRIC                                                |                                                                                                                                                                                           |                                                                             |                                                                                    |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Criterion                                             | Excellent (4)                                                                                                                                                                             | Proficient (3)                                                              | Developing (1–2)                                                                   |
| Correct use of characteristics                        | Uses cell type, number of cells, cell wall, chlorophyll and mode of nutrition accurately to place organisms in the five kingdoms and explains why they are better than personal criteria. | Uses most characteristics correctly with a few gaps in explanation.         | Uses few or incorrect characteristics, or relies on colour, habitat or usefulness. |
| Evidence and justification (Claim-Evidence-Reasoning) | Gives a clear claim supported by specific evidence from the lessons and reasoning that links the evidence to the claim and to the Lesson 1 puzzle.                                        | Gives a claim with some evidence, but the reasoning is partial.             | Gives a claim with little or no evidence or reasoning.                             |
| Accuracy of classification and naming                 | Orders the seven ranks correctly and writes scientific names with the correct format and capitalisation, italics or underlining.                                                          | Mostly correct ranks and names with minor format errors.                    | Several errors in rank order, kingdoms or the format of scientific names.          |
| Communication and collaboration                       | Writes clearly and logically using correct terms and shows use of group ideas and peer teaching.                                                                                          | Mostly clear, with some terms used loosely.                                 | Unclear or disorganised, with few key terms used.                                  |
| Local application and conservation connection         | Uses Kenyan organisms and places and explains how taxonomy helps identify disease vectors, pests and useful organisms and conserve biodiversity.                                          | Gives a local example or a conservation link, but with limited explanation. | Gives no local example or conservation link.                                       |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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